



**SCHOOL OF
COMPUTING**

DESIGN AND ANALYSIS OF ALGORITHMS

LAB WORKBOOK

WEEK - 7

NAME : B.SIDDARTHA REDDY

ROLL NUMBER : CH.SC.U4CSE24104

CLASS : CSE-B

Question 1: To implement the Greedy algorithm for the Job Sequencing with Deadlines problem and determine the optimal sequence of jobs that maximizes total profit.

Consider a set of 14 jobs, where each job requires one unit of time for completion. Each job has an associated profit and deadline. A job must be completed on or before its deadline to earn the profit.

The profits of the jobs are:

{22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11}

The deadlines of the jobs are:

{3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1}

CODE:

```
//CH.SC.U4CSE24104-- B.Siddartha Reddy
#include <stdio.h>
#define MAX 50
struct Job {
    int id;
    int profit;
    int deadline;
};
int main() {
    printf("CH.SC.U4CSE24104 -- B.Siddartha Reddy\n");
    int n, i, j;
    printf("Enter number of jobs: ");
    scanf("%d", &n);
    struct Job jobs[MAX];
    for(i = 0; i < n; i++) {
        jobs[i].id = i + 1;
        printf("\nEnter profit for Job %d: ", i+1);
        scanf("%d", &jobs[i].profit);
        printf("Enter deadline for Job %d: ", i+1);
        scanf("%d", &jobs[i].deadline);
    }
    for(i = 0; i < n-1; i++) {
```

```

        for(j = 0; j < n-i-1; j++) {
            if(jobs[j].profit < jobs[j+1].profit) {
                struct Job temp = jobs[j];
                jobs[j] = jobs[j+1];
                jobs[j+1] = temp;
            }
        }
    }

    int maxDeadline = 0;
    for(i = 0; i < n; i++) {
        if(jobs[i].deadline > maxDeadline)
            maxDeadline = jobs[i].deadline;
    }

    int slot[MAX];
    int totalProfit = 0;
    for(i = 0; i <= maxDeadline; i++)
        slot[i] = -1;
    for(i = 0; i < n; i++) {
        for(j = jobs[i].deadline; j > 0; j--) {
            if(slot[j] == -1) {
                slot[j] = jobs[i].id;
                totalProfit += jobs[i].profit;
                break;
            }
        }
    }

    printf("\n\nTime Slot\tJob\n");
    for(i = 1; i <= maxDeadline; i++) {
        if(slot[i] != -1)
            printf("%d\t\tJ%d\n", i, slot[i]);
        else
            printf("%d\t\t--\n", i);
    }
}

```

```
printf("\nMaximum Profit = %d\n", totalProfit);  
return 0;  
}
```

OUTPUT:

```
D:\VAVV CHENNAI\Semester 4\Design and Analysis of Algorithms\Lab Activities\Week 7  
D:\VAVV CHENNAI\Semester 4\Job Activities\Week 7-gcc  
CH.SC.U44CSE24104 - B. Siddartha Reddy  
Enter number of Jobs: 14  
Enter profit for Job 1: 22  
Enter deadline for Job 1: 3  
Enter profit for Job 2: 19  
Enter deadline for Job 2: 1  
Enter profit for Job 3: 29  
Enter deadline for Job 3: 7  
Enter profit for Job 4: 28  
Enter deadline for Job 5: 5  
Enter profit for Job 5: 21  
Enter deadline for Job 6: 6  
Enter profit for Job 6: 17  
Enter deadline for Job 7: 13  
Enter profit for Job 9: 28  
Enter deadline for Job 9: 12  
Enter profit for Job 11: 11  
Enter deadline for Job 11: 13  
Enter profit for Job 11: 13  
Maximum Profit = 292  
  
Time Slot    Job  
1            J6  
2            J2  
3            J12  
4            J1  
5            J9  
6            J4  
7            J8  
8            J5  
9            --  
10           J10  
11           J11  
12           J13  
13           J3
```

```
Time Slot      Job
1              J6
2              J12
3              J1
4              J8
5              J9
6              J4
7              J5
8              J3
9              --
10             J7
11             --
12             J10
13             J11
14             J13

Maximum Profit = 292
```

WORKING:

Let there be 14 jobs with the Profit of i) 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 29, 19, 11.

ii) 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1

2) DATA ANALYTICS AND INTELLIGENCE LABORATORY.

LAB-7

Job Sequencing

To implement the greedy algorithm for the job sequencing with deadline and determine the optimal sequence of jobs that maximises the total Profit.

Consider a set of 14 jobs, requires one unit of time for completion. Each job has an associated Profit and deadline.

The Profit of the jobs are ----

{ 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 29, 19, 11 }

The deadline for the jobs are

~~{ 22, 19 }~~ { 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1 }

STEP-1: Arrange the jobs in descending order of Profit.

JOB	PROFIT	DEADLINE
J ₅	30	7
J ₃	29	8
J ₄	28	6
J ₇	27	10
J ₁₂	27	2
J ₁₀	26	12
J ₈	25	4
J ₉	24	6
J ₁	22	3
J ₆	21	5
J ₂	19	3
J ₁₃	19	14
J ₁₁	14	18
J ₁₄	11	1

Since the max value of deadline is 14, fourteen time slots are created

STEP-2

1) Add J₅ (30, 7)

— — — — — J₅ — — — — —

2) Add J₃ (29, 8)

— — — — — J₅ J₃ — — — — —

3) Add J₄ (28, 6)

— — — — — J₄ J₅ J₃ — — — — —

4) Add $J_7(27,10)$

— — — — — J_{11} J_5 J_3 — J_7 — — — —

5) Add $J_{12}(27,2)$

— J_{12} — — — — J_{11} J_5 J_3 — J_7 — — — —

6) Add $J_{10}(26,4)$

— J_{12} — — — — J_{11} J_5 J_3 — J_7 — J_{10} — —

7) Add $J_8(25,4)$

— J_{12} — J_8 — J_{11} J_5 J_3 — J_7 — J_{10} — —

8) Add $J_9(24,6)$

— J_{12} — J_8 J_9 J_{11} J_5 J_3 — J_7 — J_{10} — —

9) Add $J_1(22,3)$

— J_{12} J_1 J_8 J_9 J_{11} J_5 J_3 — J_7 — J_{10} — —

10) Add $J_6(21,5)$

J_6 J_{12} J_1 J_8 J_9 J_{11} J_5 J_3 — J_7 — J_{10} — —

11) Add $J_2(19,3)$

→ J_2 cannot be added in Slot-3

→ J_2 cannot be added in Slot-2

→ J_2 cannot be added in Slot-1

∴ J_2 is being skipped.

12) Add $J_{13} (19, 14)$

J_6 J_{12} J_1 J_6 J_9 J_4 J_5 J_3 — J_7 — J_{10} — J_{13}

13) Add $J_{11} (14, 3)$

J_6 J_{12} J_1 J_6 J_9 J_4 J_5 J_3 — J_7 — J_{10} J_{11} J_{13}

14) Try adding $J_{19} (11, 1)$

→ J_{19} cannot be added since Slot-1 is occupied

Final Job sequence:

J_6	J_{12}	J_1	J_6	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
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Using the greedy job sequencing algorithm, the optimal sequence of jobs yields a maximum Profit of 292.

Time Complexity: $O(N^2)$: The time complexity of the program is $O(N^2)$ because it uses bubble sort to arrange jobs by profit and a nested loop to assign jobs to time slots in the worst case.

Space Complexity: $O(N)$: The space complexity is $O(N)$ since we use arrays to store the jobs and time slots proportional to the number of jobs.